often intractable for large n

Bias can be controlled

Mini-Batch Estimation

- Computing OT between full datasets is **often intractable for large n**
- Instead, consider empirical measures built from mini-batches:
  $\hat{\alpha}_B = \frac{1}{B}\sum_{i=1}^{B}\delta_{x_i}, \quad \hat{\beta}_B = \frac{1}{B}\sum_{j=1}^{B}\delta_{y_j}$
- Use OT loss on the these empirical mini-batch distributions: $\hat{\mathscr{L}}(\theta) = S_\varepsilon(\hat{\alpha}_B(\theta), \hat{\beta}_B)$
- **Bias** from mini-batching **can be controlled** via:
  - Large enough batch size
  - Regularization parameter ε
  - Averaging over batches
- Common practice: batches of 128–1024 samples, resample at every train loop tier

Mini-Batch Estimation